

Prototyping Game Character Animations with AI in Unity

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Figure 1: Prototyping Game Character Animations with AI in Unity

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1 Abstract

Most 3D games leverage animations to breathe life into characters. Even at the early stages of game development, prototyping with working animations helps developers quickly understand and adapt the game design. Controllable characters require animations that are constrained in multiple ways in order to blend together and

avoid artifacts such as feet sliding. Authoring, modifying and iterating on such animations is a costly process that often requires high level expertise or expensive motion capture.

We present a novel AI-powered graph tool that allows users to leverage generative machine learning models, presented as nodes in the graph, and create animations from only a few high-level inputs. This greatly reduces the time and expertise required to author animations from scratch, while offering unseen levels of control with generative motion creation.

The combination of the many relevant nodes that the graph supports and the flexibility of our state-of-the-art diffusion models allows for going beyond text-to-motion and in-betweening, common in the literature. We can generate motions that can be synchronized on multiple aspects, such as duration, speed, feet contact timings, while being constrained by signals such as text-prompts and trajectories. Our diffusion models also allow for stylizing existing motions, and generating looping animations. This makes it possible, for the first time to our knowledge, to generate complete working locomotion sets from high level inputs such as text and duration, directly usable with common character controllers.

Our system being implemented in Unity, we showcase the power of this novel approach by constructing a prototype animated character with working locomotion and attack motions without pre-existing animations, in a few minutes.

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